

4	(a) waves (travels along tube) reflect at <u>closed end / end of tube</u> incident and reflected waves or these two waves are in <u>opposite directions</u> interfere or stationary wave formed if tube length equivalent to $\lambda / 4$, $3\lambda / 4$, etc.	B1	M1	A1	[3]
	(b) (i) 1. no motion (as node) / zero amplitude	B1	[1]		
	2. vibration backwards and forwards / maximum amplitude along length	B1	[1]		
	(ii) $\lambda = 330 / 880 (= 0.375 \text{ m})$ $L = 3\lambda / 4$ $L = 3 / 4 \times (0.375) = 0.28 (0.281) \text{ m}$	C1	C1	A1	[3]

5	(a) (i) $I = I_1 + I_2$	B1	[1]
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